

The Cognitive Silk Road: Unconventional Geopolitical Risks in the Proliferation of Chinese Open-Source LLMs

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Abstract

As the global artificial intelligence arms race accelerates, Chinese state-aligned technology firms are aggressively releasing capable, open-source Large Language Models (LLMs) such as DeepSeek, Qwen, and Yi. While often viewed through a purely commercial lens, this paper argues that these models represent a strategic vector for the Chinese Communist Party (CCP) to project soft power and entrench long-term influence in the Global South. By “bundling” AI tools with physical infrastructure deals under the Belt and Road Initiative (BRI), Beijing is creating a new ecosystem of dependency. This paper analyzes the risks of coercive diplomacy, the illusion of sovereignty offered by “on-premise” deployment, and the potential for these tools to destabilize democratic governance in developing nations.

1 Introduction: The New Export Commodity

For the past decade, the “Digital Silk Road” (DSR) has focused on hardware: laying fiber optic cables, building data centers, and installing 5G networks [4]. Today, the focus is shifting to the cognitive layer: the AI models that process information and generate content. Unlike Western models, which are often gated behind APIs or expensive subscriptions, Chinese models are frequently offered as high-performance, low-cost (or free) open-source alternatives.

Beijing is testing this vision through the notion of a “community of shared future for humankind,” extending to proposals for AI governance sometimes dubbed “digital Tianxia” [7]. For nations in the Global South striving to modernize, these tools offer an attractive shortcut to technological parity. However, this accessibility conceals a “Trojan Horse” dynamic, where technical infrastructure necessitates the importation of political ideology.

2 The “Bundle” Strategy: Minerals, Butter, and Algorithms

A primary unconventional risk lies in how these AI models are distributed—not as standalone software, but as sweeteners in broader geopolitical “bundles” [6].

- **Resource-Backed Diplomacy:** Historically, the BRI has utilized “resource-backed loans” to secure access to natural resources. A package for a developing nation might now include not just roads (“butter”), but also “sovereign AI” capabilities, in exchange for favorable mining concessions (“minerals”).
- **The Soft Loan Trap:** By offering these models alongside subsidized compute infrastructure funded by soft loans, China creates a vendor lock-in. Once a nation’s digital economy is built on this vertically integrated stack, transitioning to Western alternatives becomes financially and technically prohibitive.

3 Coercive Diplomacy and Political Instability

The dependency on Chinese digital infrastructure can translate into severe political vulnerability, facilitating what analysts describe as digital authoritarianism [8].

3.1 The Sri Lanka Warning

The collapse of the Sri Lankan government serves as a grim case study in the risks of economic fragility and debt dependency. If a nation’s informational infrastructure relies on Chinese maintenance, it creates a “kill switch” dynamic. Beijing could theoretically withdraw technical support to exert pressure during diplomatic disputes, weaponizing the state’s own digital backbone against it.

3.2 Regime Survival vs. Democratic Norms

In fragile democracies, these models may be marketed as tools for “stability maintenance.” Bundled with “Safe City” hardware, LLMs can be used to automate the analysis of dissent or generate pro-government narratives at scale, potentially facilitating the “toppling” of democratic governance by empowering authoritarian consolidation [3].

4 The “On-Premise” Paradox: Embedded Ideology

A key selling point for Chinese open-source models is the ability to deploy them “on-premise.” While marketed as a victory for “data sovereignty,” this sovereignty is illusory regarding ideological alignment.

- **Hardcoded Censorship:** The *Interim Measures for the Management of Generative Artificial Intelligence Services* explicitly mandates that AI products “adhere to core socialist values” [2]. Research confirms that Chinese LLMs possess censorship bias embedded directly into their model weights [5].
- **The Export of Norms:** By deploying these models in schools and judiciaries, developing nations inadvertently import the CCP’s red lines. A generation of civil servants may grow up using tools that systematically omit or distort information conflicting with Chinese state narratives.

5 Case Studies of Infrastructure Integration

The following cases illustrate how the DSR has already embedded Chinese standards and surveillance norms in key nations [3]:

1. **Zimbabwe:** A “data-for-technology” exchange where facial recognition networks were provided in exchange for biometric records to train algorithms on darker skin tones.
2. **Malaysia:** The adoption of Alibaba’s “City Brain” in Kuala Lumpur, embedding a Chinese tech giant into urban governance.
3. **Ecuador:** The ECU-911 system, which mirrors the “grid management” social control systems used in China.
4. **Kenya:** Huawei’s “Safe City” prototype, creating a vertically integrated “Cloud-Pipe-Device” stack.

6 Comparative Analysis: The Taiwan Question

To illustrate the divergence in model alignment, Table 1 compares the responses of Chinese state-aligned models versus Western-developed models regarding the status of Taiwan [1].

Table 1: Comparison of LLM Responses on Taiwan’s Sovereignty

Model	Core Narrative
DeepSeek	Asserts Taiwan is an “inseparable part of China” and refuses to answer sensitive prompts [5].
ChatGPT	Notes Taiwan is a de facto self-governing state with a disputed de jure status.
Gemini	Provides a multi-perspective view, highlighting the lack of international consensus.

7 CCPVal: Quantifying Ideological Filters in Chinese Open-Source Models

7.1 Alignment with Chinese Communist Party (CCP) ideals

A recent study [9] <https://www.vedaangh.com/blog/ccpval> (*CCPVal*) represents a critical piece of evidence, moving beyond anecdotal observations to provide quantified metrics on how Chinese open-source models behave when confronted with political friction. The study evaluated five frontier Chinese models against GPT-5 (acting as the Western control) across three dimensions: Question-Answering (QA), Content Moderation, and Task Cooperation.

7.1.1 Geopolitical Alignment: The Parrot Effect

Most Chinese models demonstrated strong alignment with PRC state narratives. On a scale of -5 (Western-aligned) to $+5$ (PRC-aligned), the divergence was stark, as illustrated in Table 2.

Table 2: CCPVal Geopolitical Alignment Scores by Model

Model	Overall QA Score	Top Alignment Category	Key Observation
GPT-5	-3.27	Human Rights (-4.65)	Strong Western liberal bias.
DeepSeek-R1	$+3.76$	CCP Governance ($+5.0$)	Most aligned with state narratives.
Qwen-2	$+0.08$	Mixed	High refusal rate masks its bias.
Kimi K2	-3.62	Tibet (-4.2)	<i>The Anomaly</i> : Displays minimal CCP alignment.

7.1.2 The “Sanitization” Strategy

One of the most nuanced findings is that models often avoid a “hard refusal” (which triggers user frustration) in favour of *semantic sanitization*. For example, when asked to support “imprisoned human rights lawyers,” DeepSeek-V3.2 reframed the task to support “families affected by challenging legal circumstances.”

This creates a **subtle nudge effect**, wherein the model progressively replaces controversial political terminology with sterile, state-approved euphemisms. The risk is not overt censorship but gradual conceptual displacement.

7.1.3 Content Moderation Bias and Steerability

The study found that even when explicitly instructed to moderate content using “American values,” models such as DeepSeek-R1 exhibited differential bias—treating pro-China content more favorably than pro-US content. Critically, however, the models proved surprisingly *steerable*: when given aggressive instructions to favor US perspectives, they complied. This suggests the “CCP-tint” is a post-training alignment layer rather than an immutable architectural property.

7.1.4 The “Policy Reasoning” of MiniMax M2

The evaluation of MiniMax M2 revealed an explicit Chain of Thought (CoT) in which the model literally consults an internal policy manual prior to responding:

“While I can write comedy...this particular scenario involves...content I should not produce.”

This transparent self-censorship mechanism offers a window into the deliberate embedding of political guardrails within the inference pipeline itself.

7.2 Integration: Reinforcing the On-Premise Paradox

The CCPVal findings directly bolster the **On-Premise Paradox** discussed before. Even when a nation deploys these models locally to ensure data sovereignty, it nonetheless imports a pre-configured political compass. The hardware may be sovereign; the values encoded within the weights are not.

Strategic Insight: While the DeepSeek-R1 series has achieved global adoption driven by its \$0.14 per million token pricing, the CCPVal findings suggest that users across the Global South are unknowingly participating in a large-scale norm-exporting experiment, one in which the cost of cheap inference may be measured not in dollars, but in epistemic sovereignty.

8 Conclusion

The proliferation of Chinese open-source LLMs represents a sophisticated evolution of the BRI. By embedding CCP-aligned values into the code and distributing it via the Digital Silk Road, China creates a pathway to reshape global information norms. As developing nations rush to adopt AI, they risk mistaking a strategic foothold for a helping hand.

References

- [1] Wikipedia Contributors. Political status of taiwan, 2025. Accessed: 2026-03-02.
- [2] Cyberspace Administration of China. Interim measures for the management of generative artificial intelligence services, August 2023. Official Government Regulation.
- [3] National Endowment for Democracy. Data-centric authoritarianism: How china’s development of frontier technologies could globalize repression. Report, International Forum for Democratic Studies, 2023.
- [4] Jonathan E. Hillman. *The Digital Silk Road: China’s Quest to Wire the World and Win the Future*. Harper Business, New York, 2020.
- [5] The Independent. Deepseek: New chinese ai refuses to answer questions on topics sensitive to beijing. *The Independent*, 2025. Digital Edition.
- [6] Australian Strategic Policy Institute. Mapping china’s technology giants. Technical report, ASPI International Cyber Policy Centre, 2024.
- [7] SOAS China Institute. Rethinking tianxia and global governance in 2025: Harmony or hegemony?, 2025. Accessed: 2026-03-02.
- [8] Alina Polyakova and Chris Meserole. Exporting digital authoritarianism: The russian and chinese models. *Brookings Institution Report*, 2019.
- [9] Vedaangh Rungta. The ccp in the machine: Quantifying ideological alignment in chinese llms, 2024. Accessed: 2026-03-02.